Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/13/2019
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Crystal-to-Glass Transitions in Colloids (Flight Experiment)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	12/19/2018	**End Date:**	12/18/2020
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0348		
Performance Goal No.:			
Performance Goal Text:			
	This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3) (Ed. note: see "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)"; grant NNX13AL27G ; Principal Investigator Arjun Yodh, in Task Book). The roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; the science goals and experimental plans for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), were also critically reviewed. The current research will be integrated into the hardware developed for the spaceflights that are named by variants of the term “ACE” (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid		

Task Description:	Alloy Test (BCAT) BCAT-series flight experiments. In response to these developments, the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the European Space Agency (ESA) and NASA facilities. The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape, and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments. This version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer, or a micelle). Phase 3 studies retained many of the original themes but also had new focus that reflected the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming, and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales. International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.
Task Progress:	In several papers published in the past year, we found: Spatiotemporal dynamics of short- and long-time structural relaxation are measured experimentally as a function of packing fraction, f, in quasi-two-dimensional colloidal supercooled liquids and glasses. The relaxation times associated with long-time dynamic heterogeneity and short-time intracage motion are found to be strongly correlated to and grow by orders of magnitude with increasing f toward dynamic arrest. We find that clusters of fast particles on the two timescales often overlap, and, interestingly, the distribution of minimum-spatial-separation between closest non-overlapping clusters across the two timescales is revealed to be exponential with a decay length that increases with f. In total, the experimental observations suggest short-time relaxation events are very often precursors to heterogeneous relaxation at longer timescales in glassy materials. We experimentally characterize heterogeneous nonexponential relaxation in bidisperse supercooled colloidal liquids utilizing a recent concept called “softness” [Cubuk ED et al., Phys. Rev. Lett. 114, 108001 (2015)]. Particle trajectory and structure data enable classification of particles into subgroups with different local environments and propensities to hop. We determine residence times tR between particle hops and show that tR derived from particles in the same softness subgroup are exponentially distributed. Using the mean residence time tR for each softness subgroup, and a Kramers’ reaction rate model, we estimate the activation energy barriers Eb for particle hops, and show that both tR and Eb are monotonic functions of softness. Finally, we derive information about the combinations of large and small particle neighbors that determine particle softness, and we explicitly show that multiple exponential relaxation channels in the supercooled liquid give rise to its nonexponential behavior. Liquid structure and dynamics are experimentally investigated in colloidal suspensions with short-range depletion attraction. The colloidal fluid samples consist of hard-sphere colloidal particles suspended along with rodlike depletants based on surfactant micelles. The spheres have a range of surface chemistries, diameters, and packing fractions, and the rodlike micelle length depends on the temperature. Thus, the combination of hard-spheres and depletants generates a sample wherein short-range interparticle attraction can be temperature-tuned in situ. Video optical microscopy and particle tracking techniques are employed to measure particle trajectories from which structural and dynamical quantities are derived, including the particle pair correlation function [g(r)], mean square displacement, long-time diffusion coefficient, and the sample two-body excess entropy (S2). The samples with stronger short-range attractions exhibit more order, as characterized by g(r) and S2. The stronger short-range attractions are also observed to lead to slower long-time diffusion and more heterogeneous dynamics at intermediate time scales. Finally, the excess entropy scaling law prediction, i.e., the exponential relationship between two-body excess entropy and long-time diffusivity, is observed across the full range of samples. The dynamics in supercooled liquids slow enormously upon approaching the glass transition, albeit without significant change of liquid structure. This empirical observation has stimulated development of many theoretical models which attempt to elucidate microscopic mechanisms in glasses and glass precursors. Here, quasi-two-dimensional colloidal supercooled liquids and glasses are employed to experimentally test predictions of widely used models: mode coupling theory (MCT) and its important extension, inhomogeneous MCT (IMCT). We measure two-point dynamic correlation functions in the glass forming liquids to determine structural relaxation times, ta, and mode coupling exponents, a, b, and γ ; these parameters are then used to extract the mode coupling dynamic crossover packing area-fraction, fc. This information, along with our measurements of supercooled liquid spatiotemporal dynamics, permits characterization of dynamic heterogeneities in the samples and facilitates direct experimental tests of the scaling predictions of IMCT. The time scales at which dynamic heterogeneities are largest, and their spatial sizes, exhibit power law growth on approaching fc. Within experimental error, the exponents of the measured power laws are close to the predictions of IMCT.
Bibliography Type:	Description: (Last Updated: 06/16/2025)

Articles in Peer-reviewed Journals	Mishra CK, Ma X, Habdas P, Aptowicz KB, Yodh AG. "Correlations between short- and long-time relaxation in colloidal supercooled liquids and glasses." Physical Review E. 2019 Aug;100(2-1):020603. https://doi.org/10.1103/PhysRevE.100.020603 ; PubMed PMID: 31574722 , Aug-2019
Articles in Peer-reviewed Journals	Ma X, Davidson ZS, Still T, Ivancic RJS, Schoenholz SS, Liu AJ, Yodh AG. "Heterogeneous activation, local structure, and softness in supercooled colloidal liquids." Physical Review Letters. 2019 Jan 18;122(2):028001. (Editor's Choice) https://doi.org/10.1103/PhysRevLett.122.028001 ; PubMed PMID: 30720295 , Jan-2019
Articles in Peer-reviewed Journals	Ma X, Liu J, Zhang Y, Habdas P, Yodh AG. "Excess entropy and long-time diffusion in colloidal fluids with short-range interparticle attraction." Journal of Chemical Physics. 2019 Apr 14;150(14):144907. (Editor's Choice). https://doi.org/10.1063/1.5091564 ; PubMed PMID: 30981231 , Apr-2019
Articles in Peer-reviewed Journals	Mishra CK, Habdas P, Yodh AG. "Dynamic heterogeneities in colloidal supercooled liquids: experimental tests of inhomogeneous mode coupling theory." Journal of Physical Chemistry B. 2019 Jun 20;123(24):5181-8. https://doi.org/10.1021/acs.jpcb.9b03419 ; PubMed PMID: 31132279 , Jun-2019